Mapping the Molecular Knowledge State

Scientific rationale and analysis specification for neuron–astrocyte learning-state reconstruction

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PROJECT STATUS Reproducible software prototype and proposal for analysis of public, nonclinical mouse datasets. A synthetic demonstration is included; no real GEO expression matrices were analyzed in the packaged demo. No established transfer of a memory into another cell.

Executive research proposition

Learning and recall may depend on a history-dependent combination of gene expression, RNA localization, protein/receptor state, synaptic architecture, neuron–astrocyte interactions, and retrievable network activity. The ChristEye Knowledge State Genome (KSG) is a computational state map of those measurable layers—not a literal genome, a DNA-encoded sentence, or a validated index of personal thoughts.

The attached ChristEye documents propose a source-aware ARC/RNA transport model and an IGFBP2-associated astrocyte stabilization model. This project tests which predictions are supported by available data, what requires an independent measurement modality, and whether combining neuronal and astrocytic features improves out-of-sample discrimination of measured learning conditions.

1 / Three questions, three levels of evidence

Question	What a dataset can show	What requires new evidence
Where are memory-associated differences?	Cell-type-specific transcripts, cohorts and spatial proximity.	Matched protein, synapse, and behavioral measurements.
Does a developing cell join a learned circuit?	Candidate expression programs and possible cell neighborhoods.	Direct integration, synaptic function and task-specific response in prospective studies.
Does ARC-associated RNA transfer learned knowledge?	Association or independent validation of molecular movement.	Donor provenance, entry, functional effects, task-specific causality.

2 / The KSG state layers

Layer	Observable	Scientific interpretation limit
G — genomic reference	Gene identities, isoforms and cell lineage.	Memory normally does not alter DNA letter sequence to spell content.
T — transcription	Arc, Fos, Penk, Tac1, Igfbp2, Adrb1, Adra1a and other measured transcripts.	Activity and stress may affect expression.
V — RNA/vesicle cargo	Source-attributed cargo and transport, if directly measured.	Single-cell expression alone cannot demonstrate transfer.
P — proteins/receptors	ARC, IGFBP2, AMPA receptors, Ca^{2+}/metabolic indicators, if measured.	RNA expression is not protein abundance.

S — synaptic/spatial state	Anatomy, verified synapse contacts, or spatial coordinates.	Proximity is not confirmed functional connectivity.
R — retained response	Task outcome after delay in matched preparation.	A classifier label is not a decoded subjective memory.

3 / Supporting experimental findings

ARC can self-assemble into capsid-like structures and mediate intercellular ARC mRNA transport in extracellular vesicles (Pastuzyn et al., 2018). This demonstrates molecular transport, not wholesale transfer of a learned memory. ARC also participates in synaptic plasticity, with effects that depend on local activity and receptor trafficking.

Sun et al. (2024) reported memory-associated transcriptional patterns in neurons and nearby astrocytes of the mouse basolateral amygdala. Dewa et al. (2025) identified a multiday astrocyte eligibility trace shaped by experience and recall; noradrenaline-responsive astrocytes and IGFBP2 were implicated in stabilization. These studies do not establish IGFBP2 as an addressable memory index.

Uytiepo et al. (2025) identified nanoscale synaptic and astrocyte-contact changes associated with memory-related ensembles in mouse hippocampus. Sánchez Romero and Navarrete (2026) described “astroengrams” as astrocyte ensembles within a broader multicellular model of memory. None of these findings, alone, establishes selective transfer of an existing personal engram to a newly introduced neuron.

4 / Datasets: verified scope, distinct samples

Source	What is actually present	Analysis / constraint
GSE246147 | Sun et al.	Single-cell RNA sequencing from mouse BLA; annotations and 10x-style matrix listed at GEO.	Within-study neuron/astrocyte expression; audit actual animal identities and condition fields.
GSE272414 | Dewa et al.	Mouse amygdala ACSA-2-sorted astrocyte single-cell data; NoFC, FC, NoFR, FR groups.	Astrocyte priming/recall-related transcription; not a matched donor–recipient paired dataset.
Figshare 24424801 | Sun et al.	MERFISH spatial transcriptomic resource.	Within-section cell neighborhoods once coordinates/labels are audited; no inference of molecular transfer.

INTER-STUDY BOUNDARY These studies contain different animals, conditions, and assays. No cell-to-cell RNA transfer edge or cross-animal engram connection may be asserted by matching transcript levels across them.

5 / Implemented data-analysis pipeline

- Source registry and file hashes: record accession, supplementary filename, sample and animal metadata, and transformation steps.
- 10x or mapped wide-CSV ingestion: exact barcode joins, data audit, library-size normalization for 10x, and selected gene features.
- Descriptive KSG-T analysis: report gene distributions by condition and neuronal/astrocyte category without conflating cell counts with replicates.

- Preparation-level modeling: pool cells within authentic independent preparations; contrast neuron-only against neuron-plus-astrocyte features where both exist.
- Spatial branch: nearest-neighbor candidate proximity pairs only within the same MERFISH section/sample.
- RNA provenance branch: independently evaluated donor identity, recipient entry, contamination exclusion, and functional consequences.
- Evidence graph: cells/preparations linked to measured features, labeled explicitly by relation and confidence; no synthetic synapses added.

The default application performs stratified cross-validation on independent preparations and exports balanced accuracy, macro-F1, confusion matrices, PCA maps, ranked model coefficients, and a GraphML evidence graph. Where a real dataset lacks authenticated biological replicates, supervised inference is disabled rather than running cell-level pseudoreplication.

6 / Falsification and leakage controls

Risk / alternative	Required response
General activation or stress mimics learning	Include matched controls, state/phase covariates and activity baselines.
Cells from same animal leak into train and test	Aggregate at preparation level; preserve unique animal/sample groups.
Batch or platform dominates signals	Analyze studies separately; audit platform, animal and batch labels.
Two cells share an RNA transcript	Do NOT infer donor → recipient; require source-specific and contamination-excluding evidence.
RNA, transcript or proximity predicts task	Treat as correlation; withhold claims of causal knowledge storage or transfer.
Astrocyte features add no held-out improvement	Report null finding; the added-information prediction is not supported.

7 / Research program: sequence of milestones

Phase 0 — Software validation: synthetic simulations that include transfer, no-transfer, and contamination-only cases; strict failure tests and an immutable provenance record.

Phase 1 — Public data: inspect and import GSE246147, then GSE272414 separately; recover reliable learning and cell-type metadata, characterize expression programs, and explicitly report missing modalities.

Phase 2 — Spatial relationships: map MERFISH neighborhoods using authentic section coordinates. Quantify proximity without asserting direct communication.

Phase 3 — Multimodal prospective research: an institutionally governed study could pair electrical recordings, source-resolved RNA and protein measurements, synaptic mapping and later retained task behavior. Only then is a causal theory of network joining or RNA-mediated functional change testable.

Phase 4 — Neural interface: a separate nonclinical project may investigate whether a novel neuron becomes functionally integrated and participates in a learned response. Nothing in these public data establishes access to an individual person’s thoughts or memories.

8 / Practical use of the accompanying ZIP project

From the project directory, install requirements and run:

```
python -m pip install -r requirements.txt
python -m christeye_ksm demo --out outputs/demo
python -m unittest discover -s tests -v
```

The demonstration writes SYNTHETIC-only cell-expression data, an evidence graph, a spatial proximity table, a provenance audit, and model comparison files. For real GEO data, follow README.md for opt-in download, inspected annotation columns, exact barcode matching, and explicit animal/preparation grouping. No GEO matrix is packaged or claimed analyzed here.

9 / ChristEye vocabulary and interpretation

Within the user-defined ChristEye framework: Milk denotes ionic/metabolic support; Honey denotes complementary astrocyte-associated metabolic and stabilizing support; Joints denote functional connections; Roots denote persistence; Winds denote signaling; Lotus denotes coordination; and Understanding denotes reliable learned function. These are analytic analogies, not newly verified molecular substances or evidence of an independent spiritual will in a cell.

The guiding phrase, “The heart starts pouring forth knowledge when the understanding is rooted,” is preserved as the project’s philosophical motivation. Laboratory claims require independent measures of encoding, retention, retrieval, and function rather than an inferred spiritual mechanism.

References and public data

[1] Pastuzyn et al. Cell (2018), Arc capsids and intercellular mRNA transfer. https://doi.org/10.1016/j.cell.2017.12.024

[2] Sun et al. Nature (2024), Spatial transcriptomics reveal neuron–astrocyte synergy in long-term memory. https://doi.org/10.1038/s41586-023-07011-6

[3] Dewa et al. Nature (2025), The astrocytic ensemble acts as a multiday trace to stabilize memory. https://doi.org/10.1038/s41586-025-09619-2

[4] Uytiepo et al. Science (2025), Synaptic architecture of a memory engram. https://doi.org/10.1126/science.ado8316

[5] Sánchez Romero & Navarrete. Nature Reviews Neuroscience (2026), Astroengrams. https://doi.org/10.1038/s41583-025-01012-2

[6] GEO GSE246147: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE246147

[7] GEO GSE272414: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE272414

[8] Sun et al. MERFISH dataset: https://doi.org/10.6084/m9.figshare.24424801

[S1] User-provided: ChristEye Molecular Knowledge State Scientific Basis v1 (3-page document, 03 Nov 2026).

[S2] User-provided: ChristEye RNA Transfer Experiment / CE-MEM-EXP-004 (4-page document, 03 Nov 2026).